Conceptual Thinking

Domain Modeling advanced, Associative classes, Constraints & Enumeration

Karel de Grote Hogeschool

Modeling history

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In this course we have an agreement that we always want to model history.

What do we mean by history?

History in this context means that we want to “remember” within our system everything that has happened in the past. Such as the past enrolments for a student.

When we choose to model history instead of just saying “A student is enrolled in a course” we can say “A student has been enrolled in multiple courses over time”

When we model history this has an impact on

- The multiplicities
- The associations

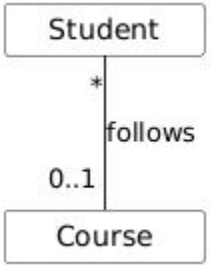
Impact on Multiplicities

When we model history, our multiplicities increase because relationships no longer have just a single value.

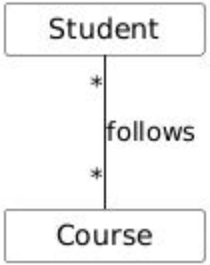
When we start modeling history, relationships that were one-to-one or one-to-many can become many-to-many relationships.

Be careful we usually resolve this by introducing an intermediate class, see the next section.

Without history



With history



Impact on Associations

When we model history, direct associations between 2 entities may disappear, being replaced by an association through an intermediate class.

In our example, if we choose to add an intermediate class “Enrolment” we may see our association change from

“Student follows a course”

to

“Student has many enrolments” and “An enrolment is linked to a course”.

Solving many-to-many associations

Solving many-to-many associations

In most cases, a many-to-many relationship should not remain in a practical class diagram design.

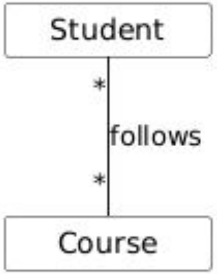
- *There are of course exceptions when it is a true many-to-many relationship.*

  An example of such a relationship is between an Author and a Book, a Book can have many authors and an Author can write many books, this is a true many-to-many and does not need splitting.

However, in most cases it is necessary to split a many-to-many relationship into an extra class.

There are 2 ways to create this extra class, it can be using an “Intermediate Class” or using an “Associative Class”.

Both are equal solutions!

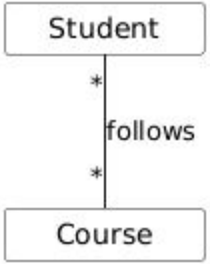


Solving many-to-many associations

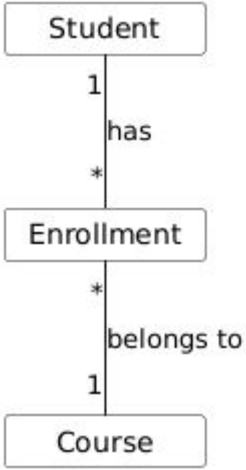
This extra class usually provides extra information to link the two classes together.

In the case of our students enrolling in courses example, if we choose to add an intermediate class “Enrolment” this may contain attributes such as enrolment date or academic year.

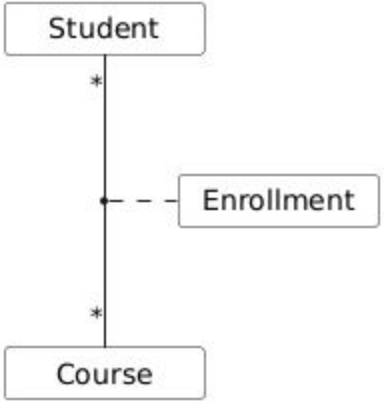
Intermediate Class **Associative Class**



Becomes



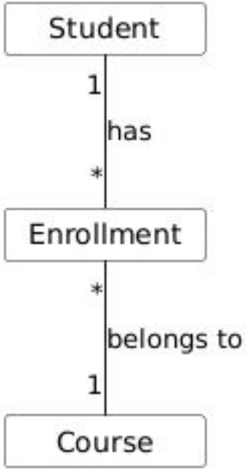
OR



Intermediate Class vs Associative Class

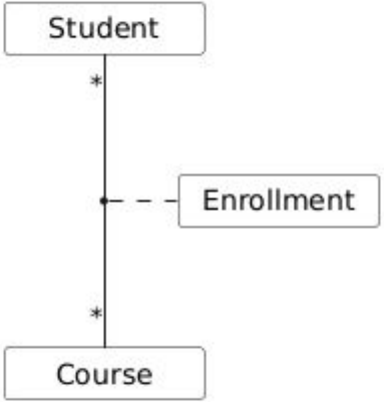
Intermediate Class

An intermediate class is a regular class that is used to split a many-to-many relationship. You model it like you would a regular class in the diagram.



Associative Class

This type of class attaches directly to an association to represent it as a class. It has the same meaning as an intermediate class but a different notation.

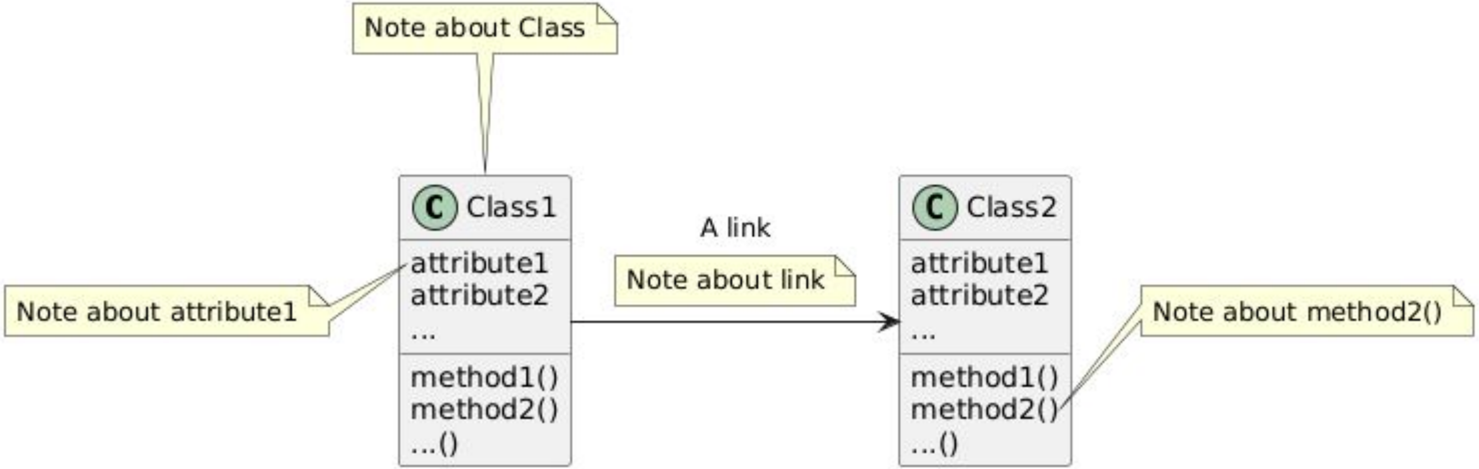


Both notations lead to equivalent solutions.

Notes and constraints

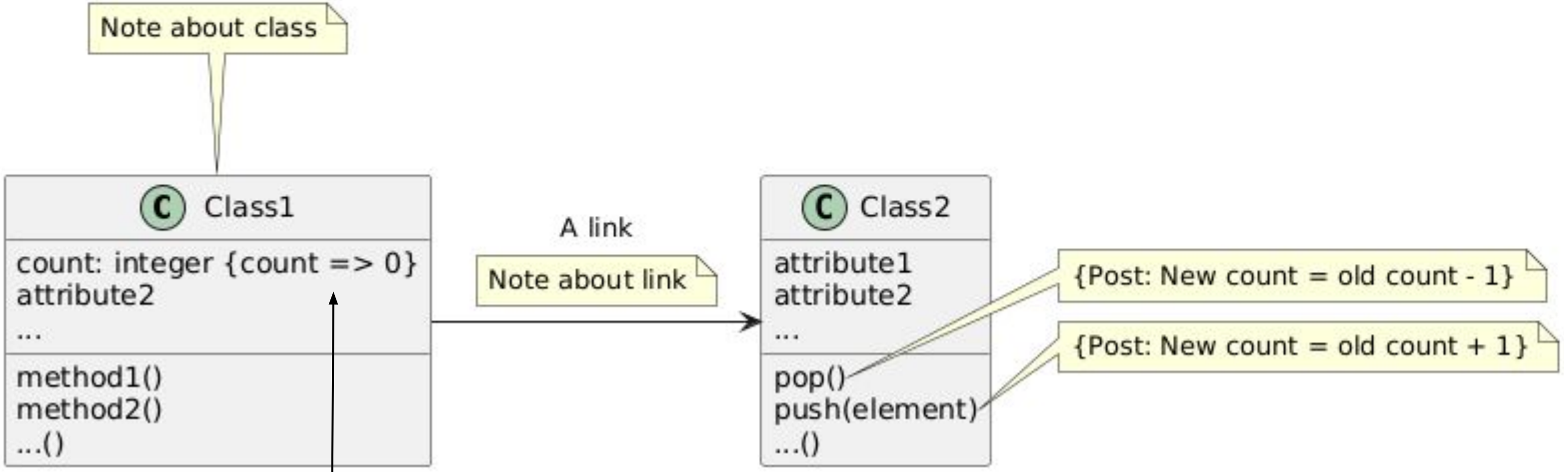
Notes

- You can add extra comments/notes to a UML diagram in a note box.



Constraints

- **Constraint = a restriction on one or more elements in the class diagram that is not "extra" or incidental but forms an essential part of the model**
  - The only rule of mention is that it is between braces {}.



In this case, we have added a constraint that count has to be a positive number

Enumeration

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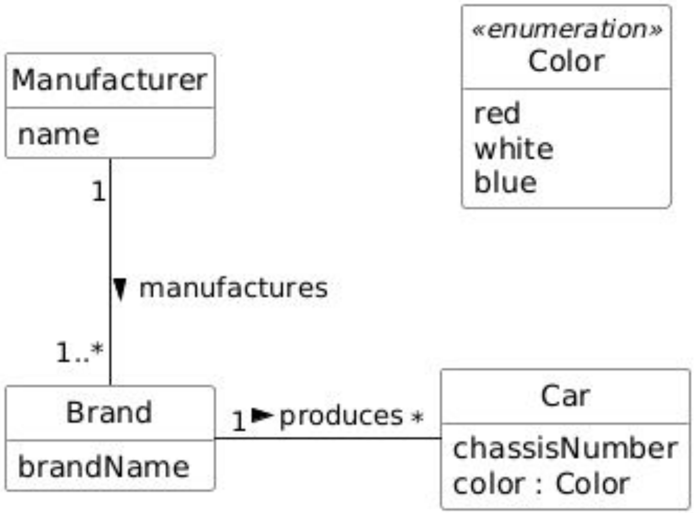
Enumeration

- Enumeration (Enum) class = A class for which we have a limited number of previously known values
  - This list of values should not be able to grow afterwards!

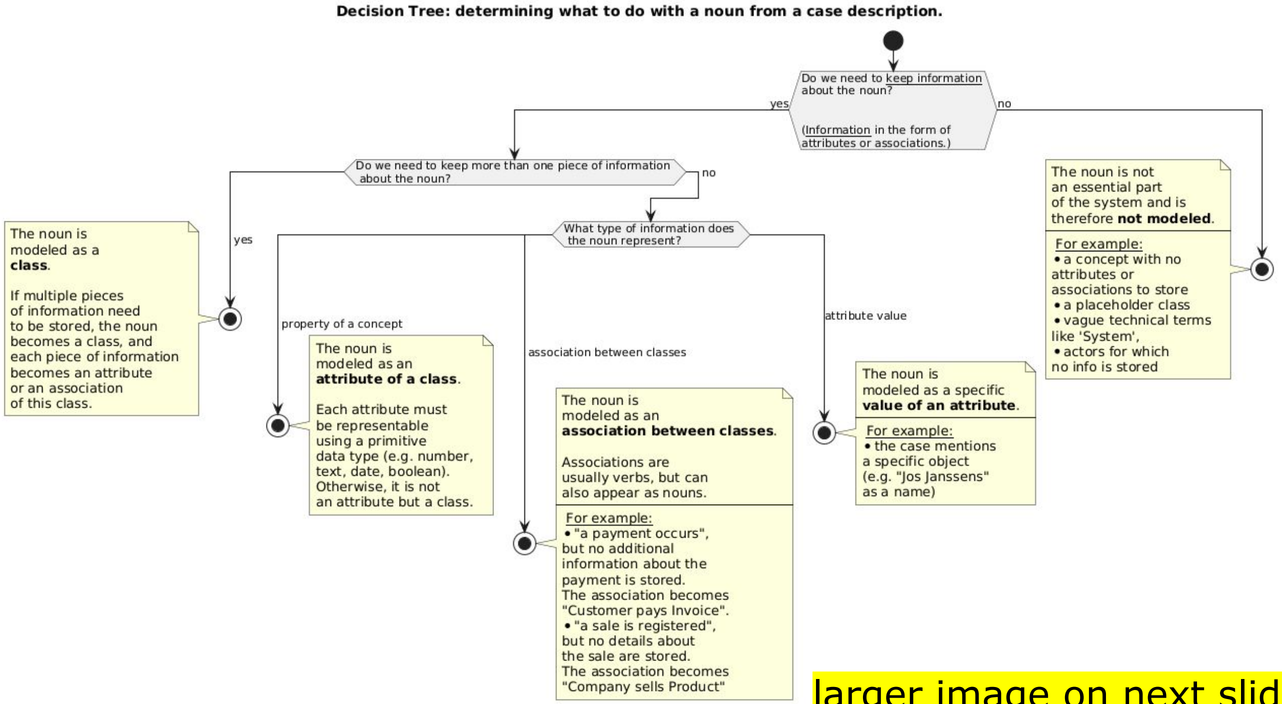
(You will learn what these look like in Java at a later time in Programming)



Decision tree noun method

Decision tree noun method

- To determine what a noun from an exercise should be in a class diagram, you can use the following decision tree:



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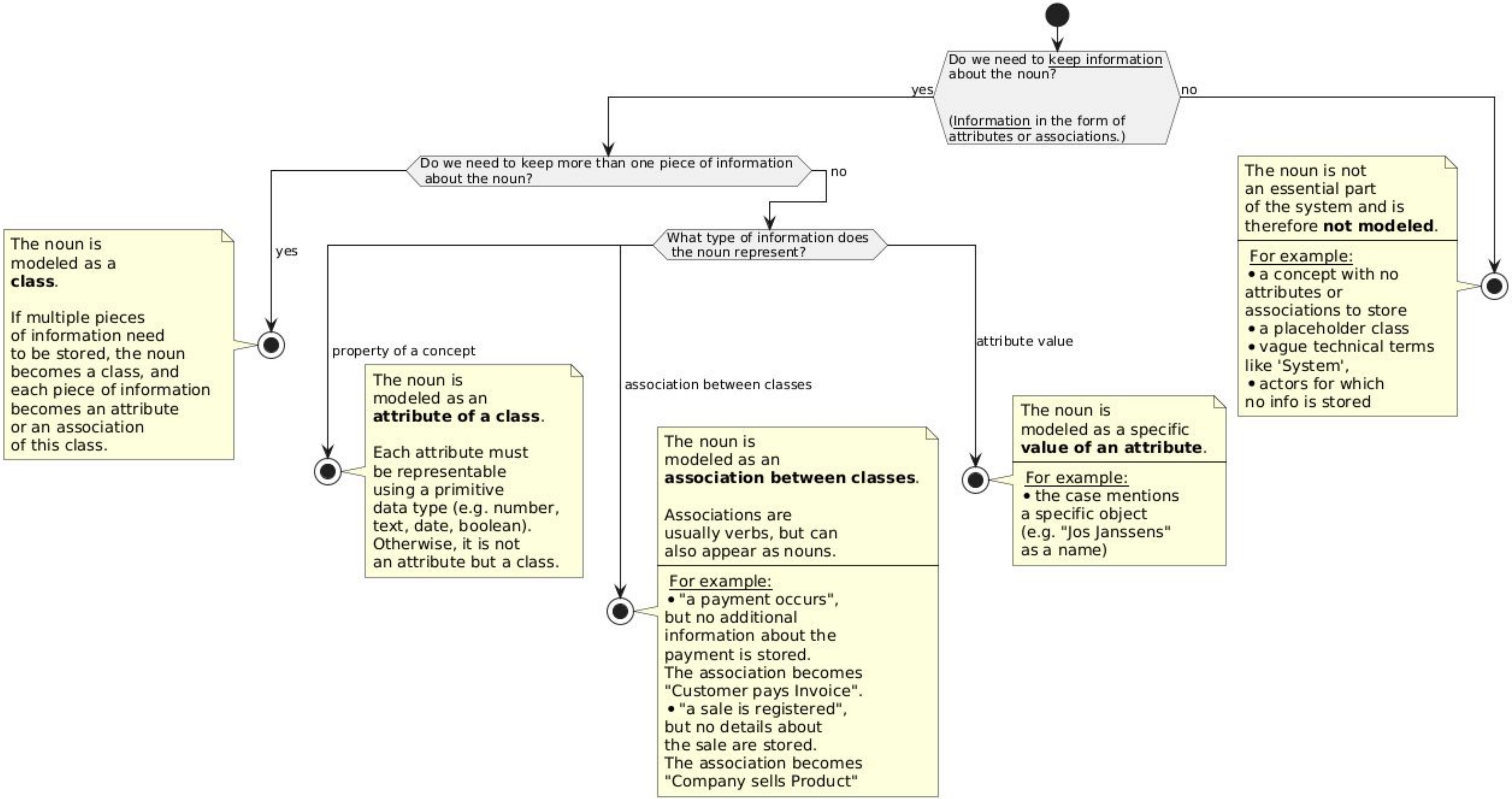
Decision Tree: determining what to do with a noun from a case description.



Sample exercise

Sample exercise

The basketball association registers all its affiliated associations (clubs) with their name and location. Club members are always assigned to a single team; each team has its own members. A team, of course, always belongs to a club. A team has a minimum of 6 and a maximum of 14 players. Each team also has one coach. The coach must also be a member of the club. A coach can, if necessary, manage multiple teams. Some teams have one or two additional trainers (also club members). Teams are designated by a group (cubs, juniors, seniors) and a ranking number. A member's name and unique membership number are known. Furthermore, each member's jersey number and date of birth are registered. This last number is necessary to determine and verify the correct group placement. The teams that play matches are assigned to a pool, whose group and division are known. Each pool contains 10 to 14 teams. A number of matches are played between two teams in each pool. Each team plays twice against every other team in its pool: once at home and once away. The date of each match is recorded. The participating team members are also recorded for each match (not every team member participates in every match). The score of each participating team member is recorded throughout the match, as well as the number of personal fouls committed. The two referees who officiated are also recorded. These two referees may not be members of either the home or away team. A referee is a member of a basketball club that holds a junior or senior license. Naturally, the license a referee holds is recorded.

Task

a. Draw a conceptual class diagram (a domain model) showing the classes with attributes, associations, and multiplicities for this system.

Sample exercise - nouns indicated

The **basketball association** registers all its affiliated **associations (clubs)** with their **name** and **location**. **Club members** are always assigned to a single **team**; each **team** has its own **members**. A **team**, of course, always belongs to a **club**. A **team** has a minimum of 6 and a maximum of 14 **players**. Each **team** also has one **coach**. *The **coach** must also be a **member** of the **club***. A **coach** can, if necessary, manage multiple **teams**. Some **teams** have one or two additional **trainers** (also **club members**). **Teams** are designated by a **group (cubs, juniors, seniors)** and a **ranking number**. A **member's name** and unique **membership number** are known. *Furthermore, each **member's jersey number** and **date of birth** are registered.* This last number is necessary to determine and verify the correct **group placement**. The **teams** that play **matches** are assigned to a **pool**, whose **group** and **division** are known. Each **pool** contains 10 to 14 **teams**. A number of **matches** are played between two teams in each **pool**. Each **team** plays twice against every other **team** in its **pool**: once **at home** and once **away**. The **date** of each **match** is recorded. The **participating team members** are also recorded for each **match** (not every **team member** participates in every **match**). The score of each **participating team member** is recorded throughout the **match**, as well as the **number of personal fouls** committed. The two **referees** who officiated are also recorded. These two **referees** may not be members of either the **home or away team**. A **referee** is a member of a **basketball club** that holds a **junior or senior license**. Naturally, the **license** a **referee** holds is recorded.

Task

a. Draw a conceptual class diagram (a domain model) showing the classes with attributes, associations, and multiplicities for this system.

Sample exercise - unique nouns indicated

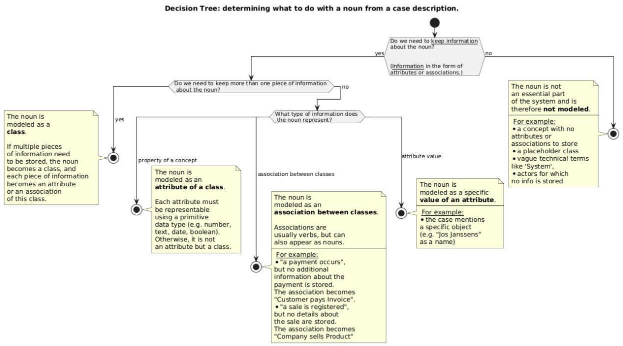
The **basketball association** registers all its affiliated **associations (clubs)** with their **name** and **location**. **Club members** are always assigned to a single **team**; each team has its own members. A team, of course, always belongs to a club. A team has a minimum of 6 and a maximum of 14 **players**. Each team also has one **coach**. *The coach must also be a member of the club.* A coach can, if necessary, manage multiple teams. Some teams have one or two additional **trainers** (also club members). Teams are designated by a **group (cubs, juniors, seniors)** and a **ranking number**. A **member's name** and unique **membership number** are known. *Furthermore, each member's **jersey number** and **date of birth** are registered.* This last number is necessary to determine and verify the correct **group placement**. The teams that play **matches** are assigned to a **pool**, whose **group** and **division** are known. Each pool contains 10 to 14 teams. A number of matches are played between two teams in each pool. Each team plays twice against every other team in its pool: once **at home** and once **away**. The **date** of each match is recorded. The **participating team members** are also recorded for each match (not every team member participates in every match). The **score** of each participating team member is recorded throughout the match, as well as the **number of personal fouls** committed. The two **referees** who officiated are also recorded. These two referees may not be members of either the home or away team. A referee is a member of a basketball club that holds a **junior or senior license**. Naturally, the license a referee holds is recorded.

Task

a. Draw a conceptual class diagram (a domain model) showing the classes with attributes, associations, and multiplicities for this system.

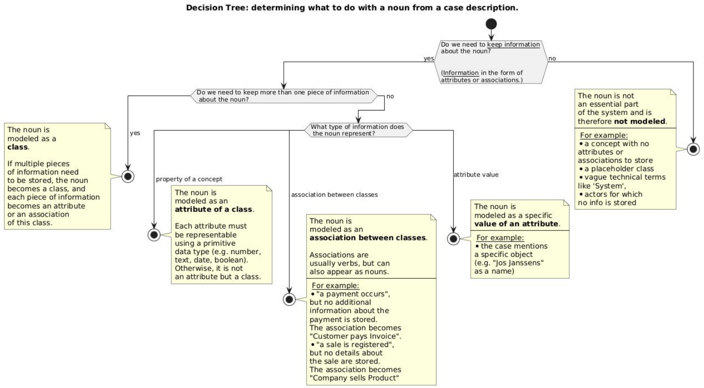
Sample exercise - applying decision tree (1)

- **basketball association** -> coat hanger class ‘Basketball Association’
- **associations (clubs)** with **name** and **location** -> class ‘Club’ with attributes
- **club members** -> class ‘Member’
- **team** -> class ‘Team’
- **players** -> class ‘Player’
- **coach** -> class ‘Coach’
- ***trainers*** -> class ‘Trainer’
- Teams are designated by a **group (cubs, juniors, seniors)** and a **ranking number** -> enumeration class ‘Group’ and attribute ranking number for class ‘Team’
- A member's **name** and unique **membership number** are known. Furthermore, each member's **jersey number** and **date of birth** are recorded. The latter is necessary to determine and verify their **placement** in the correct group. -> attributes ‘name’, ‘member number’, ‘jersey number’ and ‘date of birth’ of Member. ‘Placement’ is not modelled

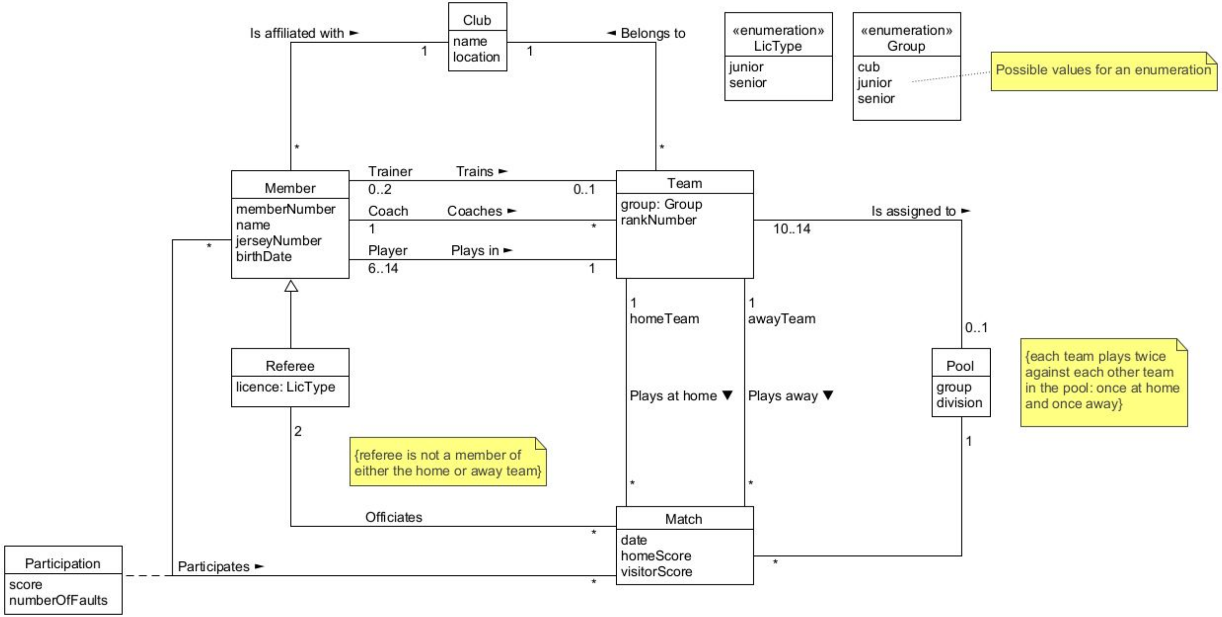


Sample exercise - applying decision tree (2)

- **matches** -> class ‘Match’
- **pool,** of which the **group** and **class** are known -> class ‘Pool’ with attributes
- Each team plays against every other team in its pool twice: once **at home** and once **away**. -> two associations between ‘Team’ and ‘Match’
- The **date** of each match is noted. The **participating team members** are also noted for each match. -> attribute ‘date’ of match and class ‘Participation’
- The **score** of each participating team member is recorded throughout the match, as well as the number of **personal fouls** committed. The two **referees** who officiated are also recorded. -> attributes ‘score’ and ‘#fouls’ of class ‘Participation’ -> association ‘Match’-‘Referee’
- A referee is a member of a basketball club who holds a **junior or senior license**. -> attribute ‘license’ of class ‘Referee’

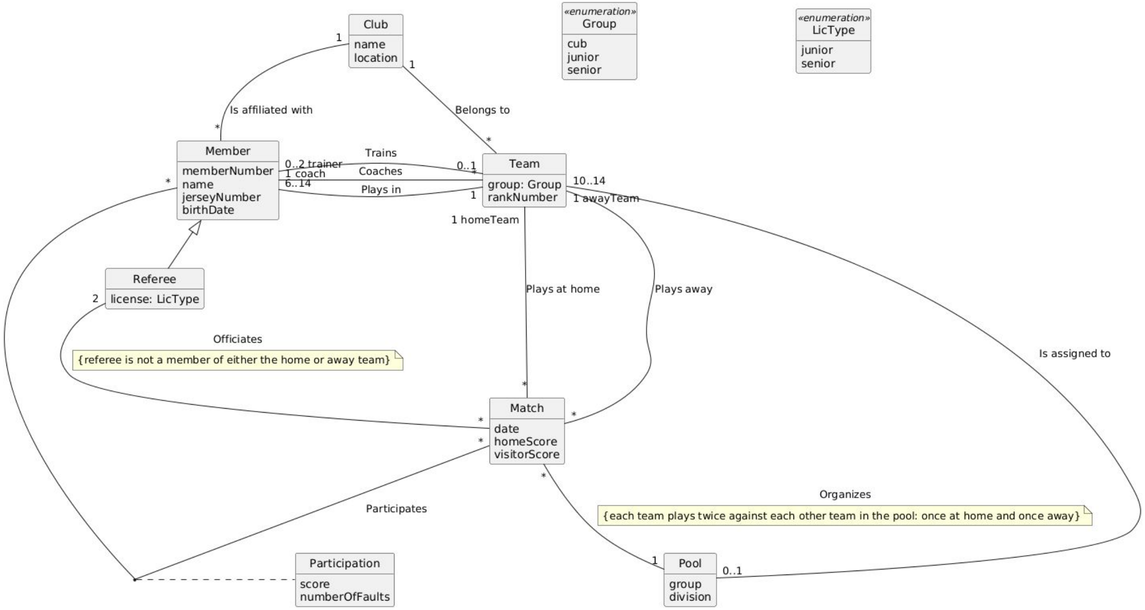


Sample exercise - model solution



- Use **enumerations** to indicate that there are defined types/kinds of a particular attribute
- Class **Participation** is essential as an associative class
- The coach and trainer are also members of the club, so keep their information in the Member class, not in a separate class.
  - Players, coaches and trainers have the same attributes
  - Also jersey number, because this wording in the assignment:
    - *"The coach must also be a member of the club"*
    - *"additional trainers (also club members)"*
    - *"Furthermore, each member's jersey number is registered."*

Sample exercise - model solution



- Use enumerations to indicate that there are defined types/kinds of a particular attribute
- Class Participation is essential as an associative class
- The coach and trainer are also members of the club, so keep their information in the Member class, not in a separate class.
  - Players, coaches and trainers have the same attributes
  - Also back number, because this wording in the assignment:
    - *"The coach must also be a member of the club"*
    - *"additional trainers (also club members)"*
    - *"Furthermore, each member's back number is registered."*

Sample exercise - Generalisation/inheritance

Inheritance can’t be used for a **Member** to **Coach**, **Trainer**, **Player**

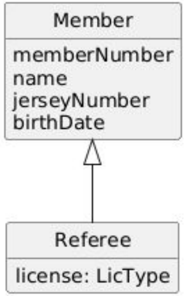
See last weeks slides

Conditions inheritance	Member to Coach, Trainer, Player
“Is-a” rule	OK
100% rule	OK
Static	NOK, the 3 members can change into a subclass
Exclusive	NOK, the 3 members can be part of more subclasses

Sample exercise - Generalisation/inheritance

Inheritance is used from **Member** to **Referee**

Conditions inheritance	**Member to Coach, Trainer, Player**
“Is-a” rule	OK
100% rule	OK
Static	OK, Referee does not change into a sub class
Exclusive	OK, Referee is part of one class



Recurring Example

Recurring Example: Monopoly

Purpose of the Exercise

This exercise is an ongoing case study in Conceptual Thinking. Each week, we apply what we've lea
of Monopoly.

Why do we choose Monopoly?

- Everyone knows it, or almost everyone
- It contains rich structures: players, money, rules, actions, spaces, cards
- It provides an ideal context for modeling object-oriented systems
- It is tangible and recognizable, making abstract concepts concrete

Recurring Example: Monopoly

Learning objective

Students understand how theory is translated into a working conceptual model by:

- Working iteratively with one consistent case
- Apply different analysis methods (use cases, class diagrams, ...)
- Gain insight into the relationship between models and system behavior

Quick Refresher: How Does Monopoly Work Again?

- 2–6 players move their pawn around the board with a roll of 2 dice
- Players buy properties they land on
- Other players pay rent when they land on someone else's property
- There are special subjects such as Chance, Community Fund, Tax, Prison, etc.
- Players can take out mortgages, trade properties, or go bankrupt
- Goal: Become the richest player by bankrupting others

Conceptual Class Diagram (refined)